Macroeconomics II Lecture Notes

Harlly Zhou

March 25, 2026

Contents

1 The Solow Model **1**

1.1 Production Function and Growth Accounting 1

1.1.1 Production Function 1

1.1.2 Growth Accounting 4

1.2 Basic Solow Model 7

1.2.1 Consumption and Investment 7

1.2.2 Graphical Analysis and Mathematical Analysis 9

1.2.3 Comparative Statics 13

1.2.4 Golden Rule 17

1.3 Augmented Solow Model 19

1.3.1 Labour Augmented Technology 20

1.3.2 Comparative Statics 24

2 Testing Solow Model and Endogenous Growth Models **27**

Chapter 1

The Solow Model

This chapter discusses long-run economic growth using a fundamental growth model: the Solow model. We start this chapter by some motivating facts from data. Then we solve for the baseline version model and study the steady state of the model. The baseline model implies no long-run growth, which motivates the augmented Solow model. We introduce the concept of balanced growth path and conclude the chapter by the result that the only way for the economy to grow over the long run is from growth in productivity and population, while productivity growth is a must for per capital variables to grow.

1.1 Production Function and Growth Accounting

In ECON 3123, we know that the size of an economy is measured by GDP, which is the aggregate output of the economy. Therefore, to analyze the long-run growth of an economy, it is important to understand how output is produced.

1.1.1 Production Function

Consider an economy where capital, denoted by K, and labour, denoted by N, are the only inputs to production, and they are mapped to output via a **production function**:

$$Y_t = A_t F(K_t, N_t), \tag{1.1}$$

where subscript t denoted the period and A_t is an **exogenous variable** which measure productivity. For simplicity, assume that $A_t \equiv A$ for all $t = 0, 1, \cdots$.

The function $F(\cdot, \cdot)$ is assumed to have the following properties.

Assumption 1.1. *Denote the first and second order partial derivatives of $F(K, N)$ by F_K, F_N, F_{KK}, F_{NN}, F_{KN}. The production function should satisfy the following conditions:*

(1) Marginal products: $F_K > 0$ *and* $F_N > 0$*;*

(2) ***Inada conditions****:*

$$\lim_{K_t \to 0} F_K(K_t, N_t) = +\infty, \quad \lim_{K_t \to +\infty} F_K(K_t, N_t) = 0,$$
$$\lim_{N_t \to 0} F_N(K_t, N_t) = +\infty, \quad \lim_{N_t \to +\infty} F_N(K_t, N_t) = 0;$$

(3) Diminishing marginal products: $F_{KK} < 0$ *and* $F_{NN} < 0$*;*

(4) $F_{KN} > 0$*;*

(5) Constant return to scale (CRTS): For $\forall \gamma > 0$*,* $F(\gamma K_t, \gamma N_t) = \gamma F(K_t, N_t)$*;*

(6) If $K_t = 0$ *or* $N_t = 0$*, then* $F(K_t, N_t) = 0$*.*

An example of such production function is the **Cobb-Douglas production function**, which we will make use of a lot.

Example 1.1 (Cobb-Douglas Production Function)**.** *Consider the Cobb-Douglas production function:*

$$F(K_t, N_t) = K_t^{\alpha} N_t^{1-\alpha} \quad \textit{with} \quad 0 < \alpha < 1. \tag{1.2}$$

The first-order partial derivatives are:

$$F_K(K_t, N_t) = \alpha K_t^{\alpha-1} N_t^{1-\alpha} = \alpha \left(\frac{K_t}{N_t}\right)^{\alpha-1}$$
$$F_N(K_t, N_t) = (1-\alpha) K_t^{\alpha} N_t^{-\alpha} = (1-\alpha) \left(\frac{K_t}{N_t}\right)^{\alpha}.$$

Since $0 < \alpha < 1$*, both* K_t *and* N_t *should be positive. Therefore, the marginal products of capital and labour are both positive. It is also easy to verify that the Inada conditions hold.*

The second-order partial derivatives are

$$F_{KK}(K_t, N_t) = \alpha(\alpha-1) K_t^{\alpha-2} N_t^{1-\alpha}$$
$$F_{NN}(K_t, N_t) = -\alpha(1-\alpha) K_t^{\alpha} N_t^{-\alpha-1}$$
$$F_{KN}(K_t, N_t) = (1-\alpha)\alpha K_t^{\alpha-1} N_t^{-\alpha}.$$

Since $0 < \alpha < 1$*,* $F_{KK} < 0$*,* $F_{NN} < 0$ *and* $F_{KN} > 0$*.*

For any $\gamma > 0$*, we have*

$$\begin{aligned} F(\gamma K_t, \gamma N_t) &= (\gamma K_t)^{\alpha} (\gamma N_t)^{1-\alpha} \\ &= \gamma^{\alpha + (1-\alpha)} K_t^{\alpha} N_t^{\alpha} \\ &= \gamma F(K_t, N_t). \end{aligned}$$

By definition, it is of constant return to scale. Suppose that the powers to capital and labour sum up to a number larger (smaller) than 1, then the Cobb-Douglas production function will be of increasing (decreasing) return to scale.

It is clear that either $K_t = 0$ or $N_t = 0$ implies that $F(K_t, N_t) = 0$.

From the functional form, we know that different combinations of capital-labour input may produce the same amount of output. Therefore, it is natural to think about to what extent the two inputs are substitutes or complements. The concept capturing this property is called the **elasticity of substitution** (EoS), denoted by ϵ. The EoS measures how responsive the capital-labour ratio K_t/N_t is to changes in the **marginal rate of technical substitution** (MRTS), which equals the ratio of marginal products F_N/F_K (the rate at which labour can be substituted for capital along an isoquant). Formally,

$$\epsilon = \frac{d \log \frac{K_t}{N_t}}{d \log \frac{F_N}{F_K}}.$$

For the Cobb-Douglas production function, we first compute the ratio of marginal products. From the derivatives above,

$$\frac{F_N}{F_K} = \frac{1-\alpha}{\alpha} \cdot \frac{K_t}{N_t}.$$

Taking logarithms: $\log \frac{F_N}{F_K} = \log \frac{1-\alpha}{\alpha} + \log \frac{K_t}{N_t}$. Therefore $d \log \frac{F_N}{F_K} = d \log \frac{K_t}{N_t}$, which implies

$$\epsilon = \frac{d \log \frac{K_t}{N_t}}{d \log \frac{F_N}{F_K}} = 1.$$

Economically, $\epsilon = 1$ means that along an isoquant, a 1% change in the MRTS (F_N/F_K) is associated with a 1% change in the capital-labour ratio: the two inputs are unit-elastic substitutes. In a competitive market where $w/R = F_N/F_K$, a 1% increase in the wage-rental ratio leads to a 1% increase in the capital-labour ratio chosen by cost-minimizing firms.

Note that the EoS of a Cobb-Douglas production function is a constant. In fact, Cobb-Douglas production functions form a subset of a family of production functions called the **constant-elasticity-of-substitution (CES) production function**. In the following exercise, you will show that the CES production function also exhibits the required properties and has constant EoS and that the Cobb-Douglas production function is a special case of the CES production function.

Exercise 1.1 (CES production function)**.** A CES production function is of the following form:

$$F(K_t, N_t) = \left[\alpha^{\frac{1}{\xi}} K_t^{1-\frac{1}{\xi}} + (1-\alpha)^{\frac{1}{\xi}} N_t^{1-\frac{1}{\xi}}\right]^{\frac{1}{1-\frac{1}{\xi}}} \quad \text{with} \quad 0 < \alpha < 1 \text{ and } \xi > 0. \tag{1.3}$$

(1) Show that the CES production function satisfies the properties in Assumption 1.1.

(2) Show that under the CES production function, the elasticity of substitution is ξ.

(3) Show that as $\xi \to 1$, $F(K_t, N_t) \to K_t^{\alpha} N_t^{1-\alpha}$.

A representative firm faces a profit maximizing problem. Denote the marginal labour cost (*i.e.*, the real wage paid to labour) as w_t and the marginal capital cost (*i.e.*, the real return to capital) as R_t. Normalizing the output price to be 1, we get the problem:

$$\max_{K_t, N_t} \Pi_t = AF(K_t, N_t) - w_t N_t - R_t K_t. \tag{1.4}$$

The first-order condition for the representative firm is:

$$w_t = AF_N(K_t, N_t) \tag{1.5}$$

$$R_t = AF_K(K_t, N_t). \tag{1.6}$$

These conditions say that the firm ought to hire capital and labour up until the point where the marginal product of the factor equals the marginal cost. The next exercise asks you to show that in this competitive market, the profit of the firm should be zero. This is known as **Euler's Theorem**.

Theorem 1.1 (Euler's Theorem)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is a homogeneous function of degree $k \geq 0$, and continuously differentiable in some open subset of $\mathbb{R}^n$, then it satisfies the following differential equation:*

$$kf(x_1, x_2, \cdots, x_n) = \sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i}(x_1, x_2, \cdots, x_n). \tag{1.7}$$

Corollary 1.2 (Partial Derivative of a Homogeneous Function)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is a homogeneous function of degree $k \geq 0$, then the partial derivative of f with respect to x_i is a homogeneous function of degree $k - 1$.*

Exercise 1.2 (Euler's Theorem with two variables)**.** Assume that the production function $F(K_t, N_t)$ admits the properties in Assumption 1.1.

(1) Show that the solution to the optimization problem (1.4) leads to zero profit. [Hint: Consider the CRTS property. Try some differentiation so that you can get a form of $F_N N_t$ and $F_K K_t$.]

(2) Show that for any $\gamma > 0$, $F_K(\gamma K_t, \gamma N_t) = F_K(F_t, N_t)$ and $F_N(\gamma K_t, \gamma N_t) = F_N(F_t, N_t)$.

1.1.2 Growth Accounting

Given the production functions, we know that output growth can potentially come from the inputs. Specifically, with Cobb-Douglas production function

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha},$$

we know that the output can grow due to productivity growth, capital growth, and labour growth. How to explicitly express the growth rate of output?

Note that the growth rate of output is given by

$$g_Y = \frac{Y_t - Y_{t-1}}{Y_{t-1}} = \frac{Y_t}{Y_{t-1}} - 1. \tag{1.8}$$

By the Cobb-Douglas production function,

$$\begin{aligned}\frac{Y_t}{Y_{t-1}} &= \frac{A_t K_t^\alpha N_t^{1-\alpha}}{A_{t-1} K_{t-1}^\alpha N_{t-1}^{1-\alpha}} \\ &= \frac{A_t}{A_{t-1}} \left(\frac{K_t}{K_{t-1}}\right)^\alpha \left(\frac{N_t}{N_{t-1}}\right)^{1-\alpha} \\ &= (1+g_A)(1+g_K)^\alpha (1+g_N)^{1-\alpha}.\end{aligned}$$

Multiplication seems not transparent enough as a decomposition and we would like to change them to additions. To do so, we take logarithms on both sides of the equation:

$$\log(1+g_Y) = \log(1+g_A) + \alpha \log(1+g_K) + (1-\alpha)\log(1+g_N).$$

To get rid of the annoying "1+" term, we can use the following approximation by Taylor's theorem:

$$\log(1+x) \approx x \quad \text{for } x \text{ small.}$$

Therefore, we get

$$g_Y \approx g_A + \alpha g_K + (1-\alpha) g_N.$$

This is the growth **accounting equation**. It says that the growth rate of output is the sum of the growth rates of productivity, capital, and labour.

Task 1.1 (Capital share and labour share (I))**.** In this question, we will use the **Penn World Table** (PWT) dataset `https://www.rug.nl/ggdc/productivity/pwt/?lang=en`. The PWT is a cross-country dataset designed for international comparisons of macroeconomic variables. It provides measures of output, capital, labor, and productivity that are comparable across countries and over time by using purchasing power parity (PPP) adjustments. You should read the tutorial note on this task before doing it.

A key feature of the PWT is its focus on real variables, such as real GDP, capital stock, and total factor productivity. These features make it widely used in growth and development research.

From the growth accounting equation, consider a Cobb–Douglas production function

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha}.$$

Taking logarithms or growth rates suggests that we may estimate α using regression. Specifically, we may consider the following specifications:

$$\log Y_t = \beta_0 + \beta_1 \log K_t + \beta_2 \log N_t + u_t, \tag{1.9}$$

$$g_Y = \gamma_0 + \gamma_1 g_K + \gamma_2 g_N + v_t, \tag{1.10}$$

where u_t and v_t are error terms.

This task asks you to examine whether these regressions can be used to estimate α.

1. **Level regression.** Using PWT data, estimate the regression in (1.9) for the United States, Japan, India, and Nigeria. Use `cgdpo` for output, `cn` for capital, and `pop` for population.

   i. Suppose the regression model is correctly specified. How would you interpret the coefficient β_0?

   ii. Report the estimated coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ for each country. Are these estimates always between 0 and 1?

   iii. Compute $\hat{\beta}_1 + \hat{\beta}_2$ for each country. Is the sum close to 1?

   iv. We now test the hypothesis

   $$H_0 : \beta_1 + \beta_2 = 1 \quad \text{vs} \quad H_1 : \beta_1 + \beta_2 \neq 1.$$

   This is a **joint hypothesis**, which can be tested using an F**-test**. A small p-value leads to rejection of H_0.

   The following table reports the test results for Japan. Based on this output, can you conclude that the sum of the coefficients is significantly different from **one**?

   Level regression: $H_0 : \ \ln K + \ln L = 1$

Model	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
Restricted	29	0.019093				
Unrestricted	28	0.013100	1	0.005993	12.808	0.001283**

   Signif. codes: *** 0.001 ** 0.01 * 0.05 · 0.1

2. **Growth regression.** Using PWT data, estimate the regression in (1.10) for the same four countries.

   i. Explain intuitively why the growth rate $g_{Y,t}$ can be approximated by $\log Y_t - \log Y_{t-1}$. A formal derivation is not required.

   ii. Suppose the regression model is correctly specified. How would you interpret the coefficient γ_0?

   iii. Report the estimated coefficients $\hat{\gamma}_1$ and $\hat{\gamma}_2$ for each country. Are these estimates always between 0 and 1?

   iv. Compute $\hat{\gamma}_1 + \hat{\gamma}_2$ for each country. Is the sum close to 1?

   v. Consider the hypothesis

   $$H_0 : \gamma_1 + \gamma_2 = 1 \quad \text{vs} \quad H_1 : \gamma_1 + \gamma_2 \neq 1.$$

Again, use an F-test. The following table reports the test results for Nigeria. Based on this output, can you conclude that the sum of the coefficients is significantly different from **one**?

Growth regression: $H_0: \ g_K + g_L = 1$

Model	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
Restricted	28	0.57109				
Unrestricted	27	0.56733	1	0.0037596	0.1789	0.6756

1.2 Basic Solow Model

Using capital and labour as inputs, firms produce more output when capital and labour increase. While labour growth is exogenous, capital growth, or **capital accumulation**, is endogenous where agents' decisions about investment determine the rate of capital accumulation. From the capital accumulation process, we can solve for the model. We can then derive where the economy will converge to in the long run. As a result, we can backward induce how household should save to maximize their consumption in the long run.

1.2.1 Consumption and Investment

From ECON 3123, we know that output in a closed economy without government spending will be either consumed or saved. The IS relationship implies that the saving is equal to investment. Mathematically, denote s as the constant saving rate. We have

$$Y_t = C_t + I_t$$
$$I_t = sY_t.$$

Where does the investment go? It is used for capital accumulation. Formally, the capital in period $t+1$ is equal to the investment in period t plus the remaining value of the capital in period t:

$$K_{t+1} = I_t + (1-\delta)K_t,$$

where $\delta \in [0,1]$ is the depreciation rate of capital. This is called the **law of motion (LoM)** of capital.

The Solow model is characterized by the following equations:

$$Y_t = AF(K_t, N_t) \tag{1.11}$$

$$Y_t = C_t + I_t \tag{1.12}$$

$$K_{t+1} = I_t + (1-\delta)K_t \tag{1.13}$$

$$I_t = sY_t \tag{1.14}$$

$$w_t = AF_N(K_t, N_t) \tag{1.15}$$

$$R_t = AF_K(K_t, N_t). \tag{1.16}$$

Having six equations with six endogenous variables (Y_t, C_t, I_t, K_{t+1}, w_t, R_t), we can solve the model.

Plugging (1.11) into (1.14) and then into the LoM (1.13), we get

$$K_{t+1} = sAF(K_t, N_t) + (1-\delta)K_t. \tag{1.17}$$

Dividing both sides by N_t, we get the LoM of capital per worker:

$$\frac{K_{t+1}}{N_t} = sA\frac{F(K_t, N_t)}{N_t} + (1-\delta)\frac{K_t}{N_t}. \tag{1.18}$$

We are almost there for per capital LoM. Note that the LHS has a time mismatch. A commonly used trick is to do the following:

$$\frac{K_{t+1}}{N_t} = \frac{K_{t+1}}{N_{t+1}}\frac{N_{t+1}}{N_t} = \frac{K_{t+1}}{N_{t+1}}(1+n),$$

where n is the population growth rate, which we for now assume to be zero. Define the per capita capital stock as $k_t = \frac{K_t}{N_t}$. We can then rewrite the production function in per capita terms:

$$y_t \equiv \frac{Y_t}{N_t} = A\frac{F(K_t, N_t)}{N_t} = AF\left(\frac{K_t}{N_t}, 1\right) = AF(k_t, 1) := Af(k_t), \tag{1.19}$$

where the red step is by the CRTS property. Now we can write the LoM of capital per capita:

$$k_{t+1} = sAf(k_t) + (1-\delta)k_t \iff \Delta k_{t+1} = sAf(k_t) - \delta k_t. \tag{1.20}$$

The last step is to express the real wage and real return to capital in terms of capital per capita. By Exercise 1.2, we have $F_K(K_t, N_t) = F_K(k_t, 1) = f'(k_t)$ and $F_N(K_t, N_t) = \frac{1}{N_t}[F(K_t, N_t) - F_K(K_t, N_t)K_t] = f(k_t) - f'(k_t)k_t$. Therefore,

$$w_t = AF_N(K_t, N_t) = A[f(k_t) - f'(k_t)k_t] \tag{1.21}$$

$$R_t = Af'(k_t). \tag{1.22}$$

Example 1.2 (Solow model with Cobb-Douglas production function)**.** *Consider the Cobb-Douglas production function:*

$$F(K_t, N_t) = K_t^{\alpha} N_t^{1-\alpha} \quad \textit{with} \quad 0 < \alpha < 1.$$

Then we have

$$f(k_t) = Ak_t^{\alpha}.$$

The model can be summarized by the following equations:

$$\begin{aligned} y_t &= Ak_t^{\alpha} \\ y_t &= c_t + \iota_t \\ k_{t+1} &= sAk_t^{\alpha} + (1-\delta)k_t \\ \iota_t &= sy_t \\ w_t &= A(1-\alpha)k_t^{\alpha} \\ R_t &= A\alpha k_t^{\alpha-1}. \end{aligned}$$

1.2.2 Graphical Analysis and Mathematical Analysis

We will start with the graphical analysis of the Solow model. As you may have already realized, lying in the centre of the model is the LoM of capital (per capita) (1.20). We would like to investigate the following two questions:

- How will k_{t+1} vary as k_t changes?
- Will the economy converge to a state $k_t = k_{t+1}$?

To answer the first question, we can consider k_{t+1} as a function of k_t and compute the derivative:

$$\frac{dk_{t+1}}{dk_t} = sAf'(k_t) + (1-\delta). \tag{1.23}$$

This provides the slope of k_{t+1} given k_t. Recall that the Inada condition imposes that

$$\lim_{k_t \to 0} f'(k_t) = +\infty, \qquad \lim_{k_t \to +\infty} f'(k_t) = 0.$$

They together implies that the slope of k_{t+1} starts with positive infinity at the origin but decreases to $1-\delta$ as k_t increases.

Figure 1.1 plots k_{t+1} as a function of k_t together with the 45-degree line along which $k_t = k_{t+1}$. The LoM curve intersect the 45-degree line at $k_t = k_{t+1} = k^*$. We refer to this point k^* as the **steady state** of the economy. A natural question to ask is whether this steady state always exsits and is unique. The following exercise asks you to show that the steady state always exists and is unique under Assumption 1.1.

Exercise 1.3 (Existence and uniqueness of steady state)**.** Suppose that Assumption 1.1 holds.

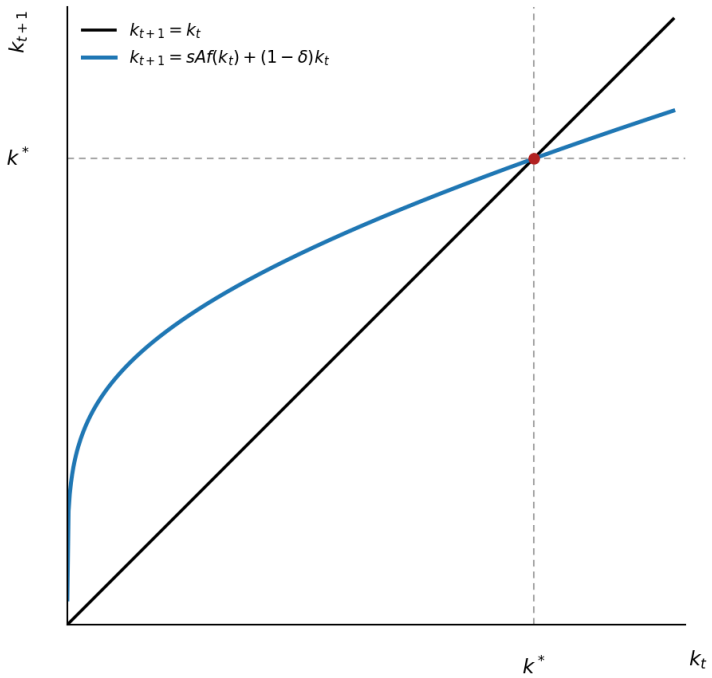

Figure 1.1: Phase diagram of the Solow model

(1) Show that there exists a steady state k^* such that $k_{t+1} = k_t = k^*$.[Hint: Consider the alternative representation of the LoM of capital per capital using Δk_{t+1}. Use intermediate value theorem.]

(2) Show that $F_{KK}(K_t, N_t) = \frac{f''(k_t)}{N_t}$. [Hin: Use Euler's theorem.]

(3) Show that the steady state is unique.

It is intuitive to conjecture that the answer to the second question is yes, and the economy will converge to the steady state. Figure 1.2 plots the convergence path of the economy from two different initial capital stocks. Starting from an initial level k_0 on the horizontal axis, move vertically to the law-of-motion curve $k_{t+1} = sAf(k_t)+(1-\delta)k_t$ to obtain k_1, then move horizontally to the 45-degree line $k_{t+1} = k_t$. Repeating these vertical and horizontal steps generates the sequence $\{k_t\}$. When $k_t < k^*$, the law-of-motion curve lies above the 45-degree line, implying $k_{t+1} > k_t$ and convergence from below. When $k_t > k^*$, the curve lies below the 45-degree line, implying $k_{t+1} < k_t$ and convergence from above. We can also see the convergence pattern if we plot the time series of k_t in Figure 1.3.

Recall that we also have an alternative representation of the LoM of capital per capita:

$$\Delta k_{t+1} = sAf(k_t) - \delta k_t.$$

This implies that depreciated capital is equal to the investment. Figure 1.4 plots the investment and depreciated capital as a function of capital per capita.

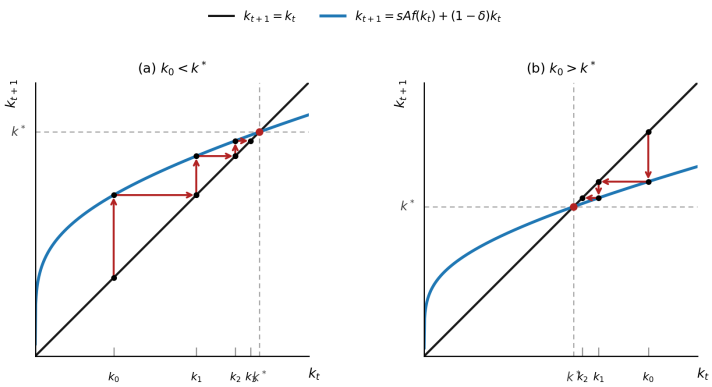

Figure 1.2: Convergence path of the Solow model

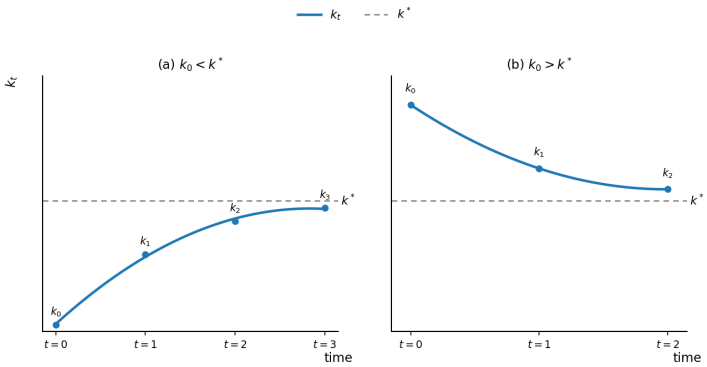

Figure 1.3: Dynamics of capital in the Solow model

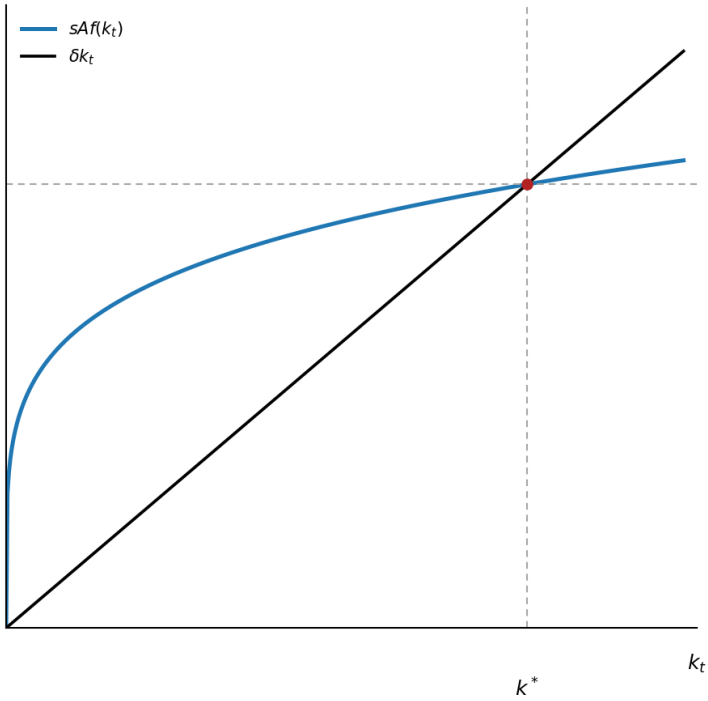

Figure 1.4: Investment and depreciated capital in the Solow model

Now we turn to the mathematical analysis of the Solow model. At the steady state, we have $k_{t+1} = k_t = k^*$. Plugging this into the LoM of capital per capita (1.20), we get

$$k^* = sAf(k^*) + (1 - \delta)k^*.$$

Rearranging the terms, we get

$$k^* = \frac{sAf(k^*)}{\delta}. \tag{1.24}$$

Alternatively, we can let $\Delta k_{t+1} = 0$ to derive the steady-state k^*. Plugging (1.24) into the system of equations that characterizes the economy, we get the following:

$$\begin{aligned}
y^* &= Af(k^*) \\
c^* &= (1 - s)y^* \\
\iota_t &= sy^* \\
w^* &= A[f(k^*) - f'(k^*)k^*] \\
R^* &= Af'(k^*).
\end{aligned}$$

Try the Cobb-Douglas specification by yourself.

1.2.3 Comparative Statics

We will now investigate how the steady state changes when the parameters change. In particular, we are interested in the change of s and A.

Change of s

Note that when s increases, (1.24) implies that k^* increases and other variables follow. This can be seen from the figure 1.5. With a positive shock to s, the LoM curve shifts upward. Given the fixed

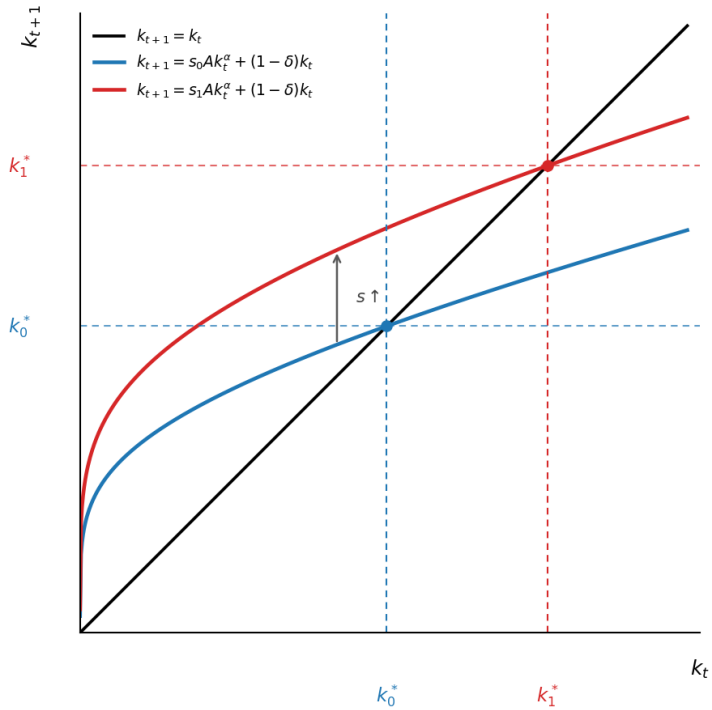

Figure 1.5: Comparative statics of s

45-degree line, the new steady state is at a higher level of capital per capita.

A more important question is how the economy transits from the old steady state to the new steady state. Figure 1.6 plots the **impulse response functions** of the economy to a positive permanent shock to s.

Before we discuss the impulse response functions, we need to clarify the concept of **stock variables** and **flow variables**. A stock variable is a variable that is measured at a point in time. It is accumulated over time. For example, capital stock K_t is a stock variable. A flow variable is a variable that is measured over a period of time. It is measured from zero when a new period starts. For example, investment I_t is a flow variable. Therefore, the value of a stock variable cannot jump because it has an accumulation process. On the other hand, the value of a flow variable can jump.

The top two panels of Figure 1.6 show the impulse response functions of k_t and y_t. As expected, k_t

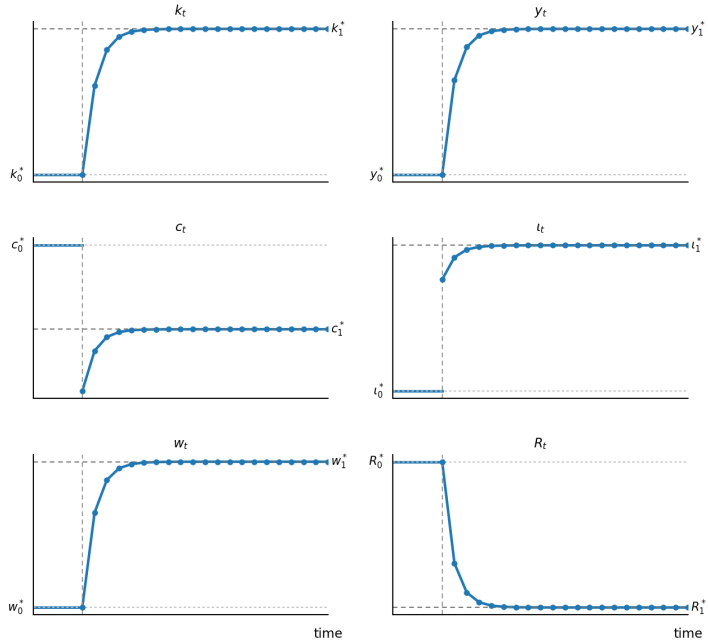

Figure 1.6: Impulse response functions to a positive permanent shock to s

and y_t increase without a jump when s increases. The effect of the shock shrinks over time, and the the two variables converge to the new steady state.

The middle to panels show the impulse response functions of c_t and ι_t. The new steady state consumption drops and the new steady state investment rises. Consider the dynamics of consumption first. Immediately after the shock, consumption drops to a level lower than the new steady state consumption. Consider the consumption equation $c_t = (1-s)y_t$. Since y_t is a stock variable, it does not immediately change, while $1-s$ is already lower. Moreover, the current $y_t = y_0^* < y_1^*$. Therefore, the concumption $c_t = (1-s_1)y_0^* < (1-s_1)y_1^* = c_1^*$. The same analysis applies to investment dynamics. Though this figure plots a lower new steady state consumption, the total effect is actually ambiguous. Consider $c^* = (1-s)y^*$. When s increases, $1-s$ decreases but y^* increases. So the effect is ambiguous.

The last two panels show the impulse response functions of w_t and R_t. The new steady state wage and return to capital both increase. The wage increases because the marginal product of capital increases. The return to capital increases because the marginal product of capital increases.

Exercise 1.4 (Decrease of s)**.** Consider an economy with a negative shock to s at time t.

(1) Draw the equilibrium diagram (like Figure 1.5). Will the new steady-state capital level increase or decrease?

(2) How will the new steady-state output, consumption, investment, wage, and return to capital change? Use math equations to explain.

(3) Draw the impulse response functions of the variables.

Change of A

Note that when A increases, (1.24) implies that k^* increases. This can be seen from the figure 1.7. While it is easy to see how the new steady-state output, consumption and investment change, it is

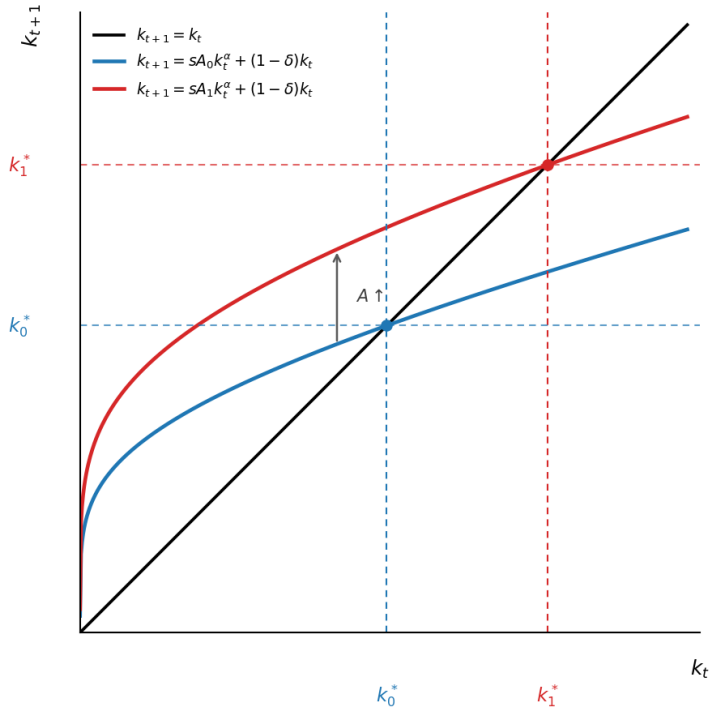

Figure 1.7: Comparative statics of A

not as straightforward to see how the new steady-state wage and return to capital change. Recall that $f''(k_t) < 0$. Therefore, an increase in k^* will leads to a decrease in $f'(k^*)$, while A increases, so the total effect is not clear, and same for w^*.

Exercise 1.5 (Change of w^* and R^*)**.** Consider an economy with a positive shock to productivity A at time t. Suppose that the production function is $Y_t = AF(K_t, N_t) = AK_t^\alpha N_t^{1-\alpha}$.

(1) Show that $f(k_t) = k_t^\alpha$.

(2) Find the expression of k^* using the parameters.

(3) Show that when A increases from A_0 to A_1, w^* increases and R^* does not change.

The impulse response functions to a positive permanent shock to A are shown in Figure 1.8.

The top two panels of Figure 1.8 show the impulse response functions of k_t and y_t. As expected, k_t and y_t increase without a jump when A increases. The effect of the shock shrinks over time, and the the two variables converge to the new steady state.

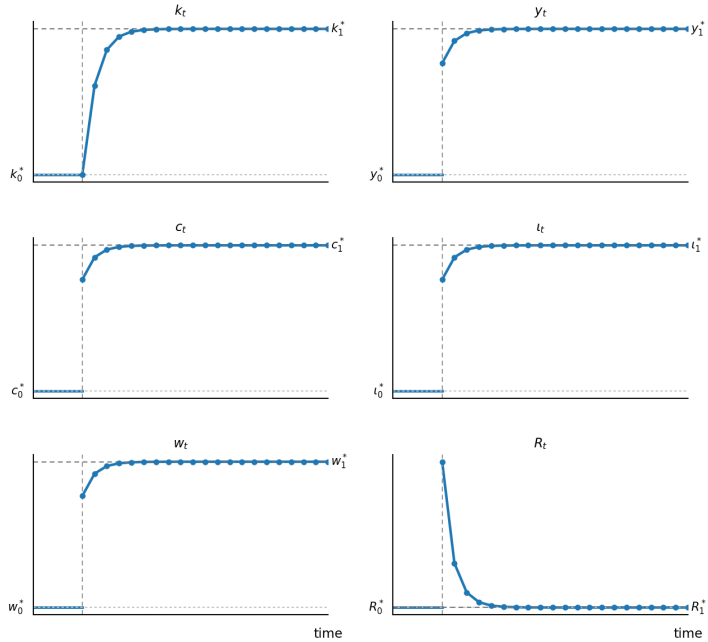

Figure 1.8: Impulse response functions to a positive permanent shock to A

The middle to panels show the impulse response functions of c_t and ι_t. The new steady state consumption and investment both increase. Again, the same reasoning applies for the jumps after the saving rate shock.

The last two panels show the impulse response functions of w_t and R_t. Different from the case of s, the two variables jumps. Note that the contemporaneous real wage is lower than the new steady state real wage while the contemporaneous return to capital is higher than the new steady state return to capital. How to understand this? Recall that the steady state real wage and return to capital are given by $w_t = A[f(k_t) - f'(k_t)k_t]$ and $R_t = Af'(k_t)$. Since k^* does not change immediately, the two variables jumps to $w_t = A_1[f(k_0^*) - f'(k_0^*)k_0^*] > A_0[f(k_0^*) - f'(k_0^*)k_0^*] = w_0^*$ and $R_t = A_1 f'(k_0^*) > A_0 f'(k_0^*) = R_0^*$. Now consider the change of w_t and R_t after the shock. Note that

$$\frac{\partial w_t}{\partial k_t} = -f''(k_t)k_t > 0, \qquad \text{and} \qquad \frac{\partial R_t}{\partial k_t} = Af'(k_t) < 0.$$

Therefore, as capital gradually increases, the real wage increases to the new steady state while the return to capital decreases to the new steady state.

Exercise 1.6 (Increase of capital depreciation rate)**.** Consider an economy with a positive shock to capital depreciation rate δ at time t.

(1) Draw the equilibrium diagram (like Figure 1.7). Will the new steady-state capital level increase or decrease?

(2) How will the new steady-state output, consumption, investment, wage, and return to capital change? Use math equations to explain.

(3) Draw the impulse response functions of the variables.

Exercise 1.7 (Decrease of capital income share)**.** Consider an economy with production function $Y_t = AK_t^{\alpha}N_t^{1-\alpha}$. Suppose that there is a negative shock to capital income share α at time t.

(1) Draw the equilibrium diagram (like Figure 1.7). Will the new steady-state capital level increase or decrease?

(2) How will the new steady-state output, consumption, investment, wage, and return to capital change? Use math equations to explain.

(3) Draw the impulse response functions of the variables.

1.2.4 Golden Rule

As we discussed in the effect of an increase in saving rate, the effect on consumption is indeterminate. However, it is just the two counterforces that make it possible to generate an optimality condition for the saving rate, that is, the saving rate that maximizes the steady state consumption, called the **golden rule** of saving rate, denoted by s_{GR}.

Consider the steady-state consumption $c^* = (1-s)y^*$. Expanding the terms, we get

$$c^* = (1-s)Af(k^*),$$

where k^* is also a function of s. Taking total derivative with respect to s, we get

$$\frac{dc^*}{ds} = -Af(k^*) + (1-s)Af'(k^*)\frac{\partial k^*}{partials}. \tag{1.25}$$

Recall that we have in (1.24) that

$$k^* = \frac{sAf(k^*)}{\delta}.$$

Taking total derivative with respect to s, we get

$$\frac{dk^*}{ds} = \frac{A}{\delta}\left[f(k^*) + sf'(k^*)\frac{\partial k^*}{\partial s}\right].$$

Substituting this into (1.25), we get

$$\frac{dc^*}{ds} = (Af'(k^*) - \delta)\frac{\partial k^*}{\partial s}.$$

To maximize c^*, we need to set the derivative to zero. Since $\frac{\partial k^*}{\partial s} > 0$, s has to satisfy the following condition:

$$Af'(k^*) = \delta. \tag{1.26}$$

Exercise 1.8 (Golden rule with Cobb-Douglas production function)**.** Consider an economy with the Cobb-Douglas production function $Y_t = AK_t^{\alpha} N_t^{1-\alpha}$. Show that $s_{GR} = \alpha$.

Exercise 1.9 (Population growth in Solow model)**.** Till now, we have assumed that the population is constant. Now we will relax this assumption. Suppose that the population grows at a rate n.

(1) Derive the law of motion for capital per labour.

(2) Derive the steady-state capital per labour.

(3) Derive the steady-state consumption per labour.

(4) Derive the golden-rule saving rate.

(5) Suppose that the economy is initially at the steady state. If n decreases at time t, how will the new steady state change? What are the intuitions?

(6) Draw the impulse response functions of capital per labour, output per labour, consumption per labour, and investment per labour. Also draw the impulse response functions of the real wage and the return to capital.

Exercise 1.10 (Two types of labour)**.** Suppose that there are two types of labour N_1 and N_2 in the economy. The production function is $Y_t = AK_t^{\alpha} \left(N_{1t}^{\theta} N_{2t}^{1-\theta}\right)^{1-\alpha}$ where $\alpha, \theta \in (0, 1)$. Assume that there is no population growth.

(1) Show that the production function is homogeneous of degree 1.

(2) Form the profit maximization problem for the firm choosing optimal K_t, N_{1t}, and N_{2t}. Assume that the real wage it w_{1t} and w_{2t} for the two types of labour respectively, and the real return to capital is R_t. Solve for the optimal labour demand for the two types of labour.

(3) Instead of doing the above maximization, assume that $N_{1t} = \eta_t N_t$ and $N_{2t} = (1 - \eta_t) N_t$. Form the profit maximization problem for the firm choosing optimal K_t, N_t, and η_t. Show that at optimum,

$$\frac{\eta_t}{1 - \eta_t} = \frac{\theta}{1 - \theta} \frac{w_{2t}}{w_{1t}}.$$

(4) Write out the per labour production function $y_t = A\Phi(\eta_t) f(k_t)$.

(5) Assume that at the steady state, $\eta_t = \eta^*$. Solve for the steady-state capital per labour k^*.

(6) Derive the steady-state output per labour y^*.

(7) Show that to maximize the steady state-output, $\eta^* = \theta$.

Task 1.2 (Capital share and labour share (II))**.** Consider the task 1.1. We know that the simple regression does not make much sense either economically or statistically to estimate α. This question helps you understand why and how economists actually calibrate the value of α. Still use the PWT dataset.

1. Consider the US economy. Plot the logarithm of real GDP, capital stock, and population. Do you see any similar trend?

2. Plot the real GDP growth rate, capital stock growth rate, and population growth rate. Do you see any similar comovement?

3. Instead of the level regression (1.9), add a time trend regressor t to the regression:

$$\log Y_t = \beta_0 + \beta_1 \log K_t + \beta_2 \log N_t + \beta_3 t + u_t. \tag{1.27}$$

   Report the estimated coefficients for the US, Japan, India, and Nigeria.

4. There are always persistence in the economy. Therefore, we always add lagged variables to the regression. Do the following regression:

$$g_{Y,t} = \gamma_0 + \gamma_1 g_{K,t} + \gamma_2 g_{N,t} + \gamma_3 g_{Y,t-1} + v_t. \tag{1.28}$$

   Report the estimated coefficients for the US, Japan, India, and Nigeria.

5. However, in practice, economists do not use the above specifications.

   a. Consider the production function $Y_t = AK_t^{\alpha} N_t^{1-\alpha}$. Let the real rental rate of capital be R_t. Show that

$$\alpha = \frac{R_t K_t}{Y_t}.$$

   b. How do you interpret α by the result?

   c. Using your interpretation, find the value of α for the US, Japan, India, and Nigeria using the PWT dataset. Explain how you get the values.

1.3 Augmented Solow Model

In the basic Solow model, the economy will converge to a steady state where there is no growth. Although we show that we can increase k^* by increasing s, this is not sustainable (why?). Therefore, we need to augment the model to allow for growth. We focus on the role of productivity.

1.3.1 Labour Augmented Technology

Consider the following production function:

$$Y_t = AF(K_t, Z_t N_t),$$

where Z_t is referred to as **labour augmenting productivity**. Its product with labour is referred to as **effective labour**. To distinguish from Z_t, we refer to A as the **neutral productivity** because it makes both factors more productive. We assume that $F(\cdot, \cdot)$ satisfies the properties in Assumption 1.1. We also have the following LoM of capital:

$$K_{t+1} = (1 - \delta)K_t + I_t,$$

where $I_t = sY_t$ is investment. The real rental rate of capital is given by

$$R_t = AF_K(K_t, Z_t N_t).$$

The real wage is given by

$$w_t = AZ_t F_N(K_t, Z_t N_t).$$

Why this Z_t appears outside the derivative? It is important to understand what F_N means. In particular,

$$F_N(K_t, Z_t N_t) = \frac{\partial F}{\partial Z_t N_t} \neq \frac{\partial F}{\partial N_t}.$$

This is because N in the subscript of the partial derivative is the second argument of the function F, not the variable N_t itself. Therefore, when you see F_N, it means to take the partial derivative with respect to the whole second argument of F. Then by chain rule, if you take partial derivative of F with respect to N_t to get the marginal product of labour, the Z_t term appears outside.

Now let's assume that labour growth is exogenous and given by $n \geq 0$. We normalize the initial level to be $N_0 = 1$. As a result, we have

$$N_t = (1 + n)^t.$$

We also allow the labour augemented producitivity to grow at a constant rate $z \geq 0$. We normalize the initial level to be $Z_0 = 1$. As a result, we have

$$Z_t = (1 + z)^t.$$

Now the following equations characterize the economy:

$$K_{t+1} = (1-\delta)K_t + I_t \tag{1.29}$$

$$I_t = sY_t \tag{1.30}$$

$$Y_t = AF(K_t, Z_tN_t) \tag{1.31}$$

$$Y_t = C_t + I_t \tag{1.32}$$

$$R_t = AF_K(K_t, Z_tN_t) \tag{1.33}$$

$$N_t = (1+n)^t \tag{1.34}$$

$$Z_t = (1+z)^t \tag{1.35}$$

$$w_t = AZ_tF_N(K_t, Z_tN_t) \tag{1.36}$$

We can then rewrite the LoM (1.29) as

$$K_{t+1} = sAF(K_t, Z_tN_t) + (1-\delta)K_t. \tag{1.37}$$

Instead of using the per labour variables, we use **per effective labour** variables. For an aggregate variable X_t, we define its per effective labour variable as

$$\hat{x}_t = \frac{X_t}{Z_tN_t}.$$

Then it is easy to see that

$$\hat{k}_{t+1} = \frac{1}{(1+z)(1+n)}\left[sAf(\hat{k}_t) + (1-\delta)\hat{k}_t\right]. \tag{1.38}$$

Other equations can be rewritten as

$$\begin{aligned}
\hat{\iota}_t &= s\hat{y}_t \\
\hat{y}_t &= Af(\hat{k}_t) \\
\hat{y}_t &= \hat{c}_t + \hat{\iota}_t \\
R_t &= Af(\hat{k}_t) \\
w_t &= Z_tA\left[f(\hat{k}_t) - f'(\hat{k}_t)\hat{k}_t\right]
\end{aligned}$$

You are required to derive the above equations by yourself. Most importantly, you need to understand how to get R_t and w_t.

Exercise 1.11 (Per effective labour variables)**.** Derive the real rental rate of capital and the real wage in terms of per effective labour variables.

Figure 1.9 shows the equilibrium diagram of the labour augmented Solow model. The only differences

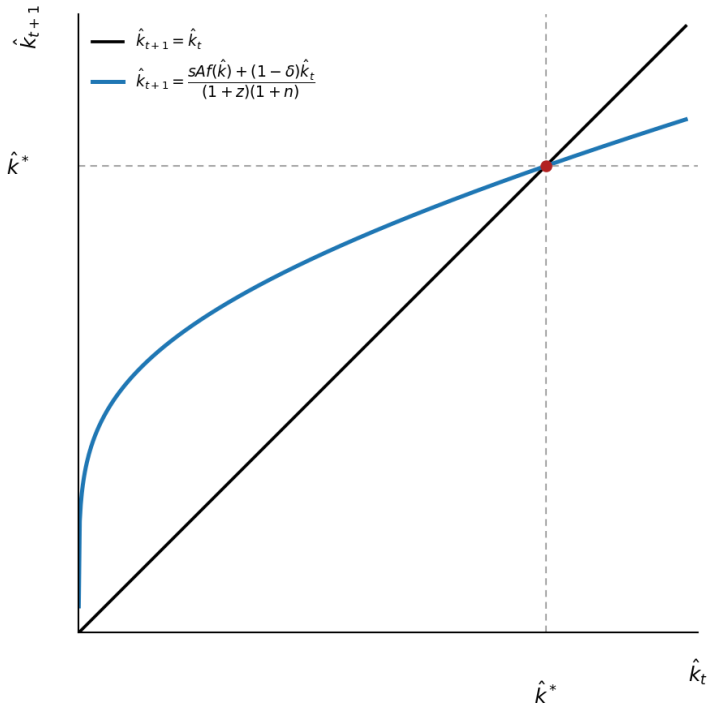

Figure 1.9: Equilibrium diagram of the labour augmented Solow model

from the basic Solow model equilibrium 1.1 are that the capital is measured in per *effective* labour term and that the LoM curve is scaled by the labour growth and labour augemnted productivity growth.

It is worth noting that people are not particularly interested in the behavior of the hat variables. They are just convenient to use. What matters to the economy are the per worker and aggregate variables. Therefore, it is natural to ask the following question: what happens to the per worker and aggregate variables when the economy converges to per effective labour variables?

At such steady state, we have $\hat{k}_{t+1} = \hat{k}_t = \hat{k}^*$. By definition, we have

$$\frac{K_{t+1}}{Z_{t+1}N_{t+1}} = \frac{K_t}{Z_t N_t},$$

which implies that

$$\frac{K_{t+1}}{K_t} = 1 + g_K = (1+z)(1+n). \tag{1.39}$$

Since z and n are typically small, we can use the log approximation to get:

$$g_K \approx z + n.$$

Similarly, we can get the growth rate of capital per worker:

$$g_k \approx z.$$

Exercise 1.12 (Aggregate variables versus per worker variables)**.** Consider the labour augmented Solow model as described by (1.29) and (1.36).

(1) Show that the growth rate of aggregate output is $g_Y \approx z + n$.

(2) Consider the following Cobb-Douglas production function:

$$Y_t = AK_t^{\alpha}(Z_t N_t)^{1-\alpha}.$$

Does the result in part (1) satisfy the growth accounting equation?

(3) Continue with part (1). Show that the growth rate of aggreagte consumption g_C and the aggregate investment I_t are equal to the growth rate of aggregate output.

(4) Let X denote any aggregate variable and x be its corresponding per worker variable. Show that $g_x = g_X - n$.

For the real rental rate of capital, since $\hat{k}_t = \hat{k}^*$, it will be constant in the long run:

$$R^* = Af'(\hat{k}^*).$$

For the real wage, consider when $\hat{k}_t \to \hat{k}^*$. We have

$$w_t = Z_t A \left[f(\hat{k}^*) - f'(\hat{k}^*)\hat{k}^* \right].$$

Therefore, we have $g_w \approx z$.

It is worth noticing that in the long run, the growth rate the the aggregate variables are constants. This is called a **balanced growth path**.

Definition 1.1 (Balanced growth path)**.** *A balanced growth path (BGP) is a path where the growth rate of all aggregate variables are constants. Mathematically, a path $\{X_t\}$ is a BGP if for each aggregate variable X, there exists a constant G_X such that*

$$\frac{X_{t+1}}{X_t} = G_X.$$

Exercise 1.13 (Twisting the production function)**.** Consider the following production function:

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha},$$

where A_t is the neutral productivity. Suppose that $A_0 = 1$ and A_t grows at a rate of $a > 0$.

(1) Show that the production function can be rewritten as

$$Y_t = \tilde{A} K_t^{\alpha}(Z_t N_t)^{1-\alpha}$$

for some constant $\tilde{A}$ and labour augmenting productivity Z_t.

(2) Suppose that the labour augmenting productivity grows at a rate of $z > 0$. Show that $z = \frac{a}{1-\alpha}$ (approximately).

1.3.2 Comparative Statics

We will now investigate how the steady state changes when the parameters change. In particular, we are interested in the change of s and A.

Change of s

After a positive shock to s, the LoM curve shifts upward in the $(\hat{k}_t, \hat{k}_{t+1})$ diagram. Given the fixed 45-degree line, the new steady state is at a higher level of $\hat{k}^*$.

What about the per labour variables? Figure 1.10 shows the impulse response functions of the economy to a positive permanent shock to s.

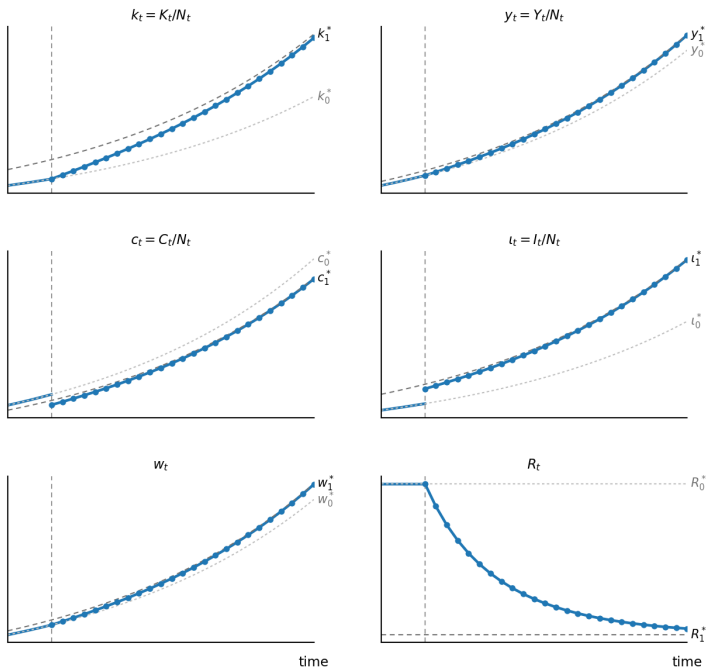

Figure 1.10: Impulse response functions to a positive permanent shock to s

We notice that except the rental rate, all variables increase even before the shock. This is because the equilibrium is a balanced growth path with their growth rate of Z_t, $z > 0$, while the rental rate does not depend on Z_t. There are also jumps and drops. the explanations are the same as in the basic Solow model.

Change of A

After a positive shock to A, the LoM curve shifts upward in the $(\hat{k}_t, \hat{k}_{t+1})$ diagram. Given the fixed 45-degree line, the new steady state is at a higher level of $\hat{k}^*$.

What about the per labour variables? Figure 1.11 shows the impulse response functions of the economy to a positive permanent shock to A. Patterns are similar to the basic Solow model.

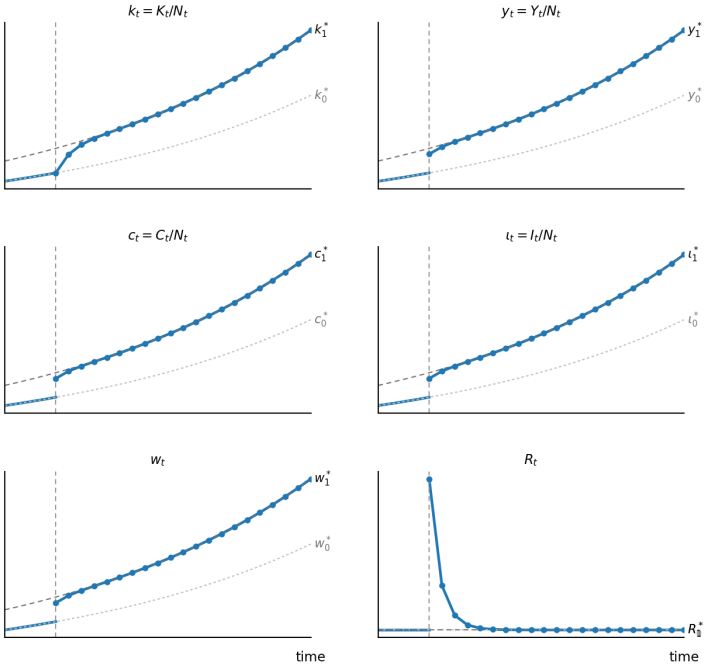

Figure 1.11: Impulse response functions to a positive permanent shock to A

Exercise 1.14 (Change of z)**.** Consider a positive shock to z.

(1) Explain intuitively how the BGP k changes.

(2) Show that the new steady state real rental rate of capital is higher.

(3) Draw the impulse response functions of the variables.

Exercise 1.15 (Human capital)**.** Consider the following production function:

$$Y_t = AF(K_t, H_t, Z_t N_t) = AK_t^{\alpha} H_t^{\beta} (Z_t N_t)^{1-\alpha-\beta},$$

where H_t is human capital and Z_t is the labour augmenting productivity. Suppose that $Z_0 = 1$ and Z_t grows at a rate of $z > 0$. Suppose that the LoMs of K_t and H_t are given by

$$K_{t+1} = s_K Y_t + (1 - \delta_K) K_t$$
$$H_{t+1} = s_H Y_t + (1 - \delta_H) H_t.$$

(1) Show that the production function $F(K_t, H_t, Z_t N_t)$ exhibits constant returns to scale.

(2) Derive for the LoMs of per effective labour capital and human capital.

(3) Compute the steady state of per effective labour capital and human capital.

(4) Compute the steady state of output per effective labour.

(5) What is the growth rate of output per labour along the BGP?

(6) Assume that $\delta_K = \delta_H = \delta$ for the remaining of the exercise. Compare this model with the augmented Solow model without human capital, that is, $F(K_t, Z_t N_t) = K_t^{\alpha}(Z_t N_t)^{1-\alpha}$. Consider an increase in s_K by θ, that is, $s'_K = s_K(1+\theta)$. Does the two model exhibit the same level of output per labour? Explain why.

Task 1.3 (Growth with Human Capital)**.** The previous exercise is based on Mankiw, Romer & Weil (1992) "*A Contribution to the Empirics of Economic Growth*", QJE, and this task asks you to read the first two sections of the paper and understand the model presented. To help you better understand hte paper, you need to know about the continuous-time version of the Solow model.

Recall that we have the following law of motion for capital:

$$\Delta K_{t+1} = s_K Y_t - \delta K_t.$$

Note that we have the time indices $t, t+1, t+2, \cdots$. In continuous time, we cannot count the time indices since the length of the time interval is super small, *i.e.*, going to zero. This matches the concept of differentiation. In your calculus class, you learned the definition of the derivative. In particular, a function $f : \mathbb{R} \to \mathbb{R}$ is differentiable at $x \in \mathbb{R}$ if

$$\frac{df}{dx} := \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

exists. If we consider the aggregate capital stock as a function of time, $K(t)$, with an time increment Δt, the LHS of the LoM becomes $K(t+\Delta t) - K(t)$. By the definition of the derivative, we have

$$\frac{dK}{dt} := \lim_{\Delta t \to 0} \frac{K(t+\Delta t) - K(t)}{\Delta t}.$$

For notation, we denote the time derivative of any variable X as $\dot{X}$. Therefore, the LoM can be rewritten as

$$\dot{K} = sY - \delta K, \tag{1.40}$$

where the time argument can be omitted.

1. By (1.40), how do you define the steady state mathematically?

2. What is the motivation for the paper to include human capital in the model?

3. How does the paper measure human capital investment? What are the potential difficulties?

4. Why the paper can conclude that human capital investment improves the model?

We will revisit this paper in the next chapter.

Chapter 2

Testing Solow Model and Endogenous Growth Models